Macroscopic materials ([redacted])

$$\begin{cases} \vec{\nabla}\cdot\vec{E} = \dfrac{\rho}{\varepsilon_0} \\ \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{E} + \dfrac{\partial\vec{B}}{\partial t} = 0 \\ \vec{\nabla}\times\vec{B} - \dfrac{1}{c^2}\dfrac{\partial\vec{E}}{\partial t} = \mu_0\vec{J} \end{cases}$$

fields → LHS
charges/currents → RHS

← true on microscopic (atomic) scale

$$\left(\frac{1}{c^2} = \varepsilon_0\mu_0\right)$$

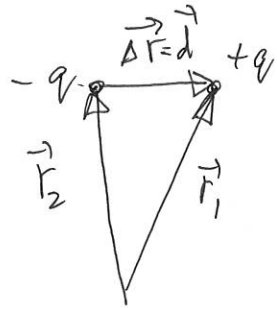
Need to average fields $\vec{E}, \vec{B}$ and $\rho, \vec{J}$ over volumes (L) large in comparison with atomic sizes but small w.r.t. physical length scale of the problem (such as wavelength, for example: $\underline{a_0 \ll L \ll \lambda}$)

Then
$$\vec{E} \to \langle\vec{E}\rangle = \vec{\mathbf{E}}$$
$$\vec{B} \to \langle\vec{B}\rangle = \vec{\mathbf{B}}$$
$$\rho \to \langle\rho\rangle = \underbrace{\rho_f}_{\text{free charges}} + \underbrace{\rho_b}_{\text{bound charges}}$$

$$\vec{J} \to \langle\vec{J}\rangle = \vec{J}_f + \vec{J}_b$$

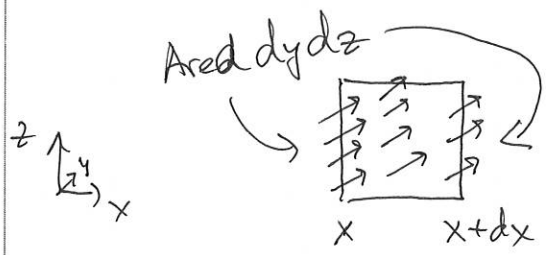
Electric polarization

$$\vec{P} = \sum_j q_j \vec{r}_j$$



$$\vec{P} = q(\vec{r}_1 - \vec{r}_2) =$$
$$= q(\vec{r}_1 - (\vec{r}_2 + \Delta\vec{r}) + \Delta\vec{r})$$
$$= q\,\Delta\vec{r}. = q\vec{d}.$$

Non-uniform polarization



$$\vec{P}(\vec{r}) = N(\vec{r})(q\vec{d})$$

left side: $N(x,y,z)(q\,d_x)\,dydz$ is charge <u>entering</u> cube via <u>face</u> $\perp$ to $\hat{x}$-axis

Right side: $N(x+dx,y,z)(q\,d_x)\,dydz$ of charge leaves the cube

$$dq = \left(-N(x+dx,y,z) + N(x,y,z)\right)(q\,d_x)\,dydz$$

$$= -\frac{\partial N(x,y,z)}{\partial x}(q\,d_x)\,dx\,dy\,dz$$

$$= -\frac{\partial P_x(x,y,z)}{\partial x}\,dV$$

$\Rightarrow$ for general orientation of $\vec{P}$ $\quad dq = -\vec{\nabla}\cdot\vec{P}\,dV$

$\Rightarrow \rho_b = -\vec{\nabla}\cdot\vec{P}.$

<u>Dielectrics</u> : no free charges / d.c. currents $\Rightarrow \langle \vec{B} \rangle = 0$

$$\Rightarrow \begin{cases} \vec{\nabla} \times \vec{E} = 0 & \left(\text{steady state} \Rightarrow \frac{\partial \vec{B}}{\partial t} = 0\right) \\ \vec{\nabla} \cdot \vec{E} = \frac{\rho_b}{\varepsilon_0} \end{cases}$$

No free charges $\Rightarrow \int \rho_b \, d^3\vec{r} = 0$ <u>always</u> (meaning: even in the presence of external $\vec{E}$)

$\Rightarrow$ can write $\rho_b = -\vec{\nabla} \cdot \vec{P}$

$\vec{P}$ = electric polarization

($\vec{P} = 0$ outside the material!)

$$\int \rho_b \, d^3\vec{r} = -\int \vec{\nabla} \cdot \vec{P} \, d^3\vec{r} = -\underbrace{\oint \vec{P} \cdot d\vec{S}}_{\substack{\text{surface} \\ \text{covering all} \\ \text{of a body}}} = 0.$$

$\vec{P}(\vec{r} \in S) = 0$

Connection with dipole moment:

$$\int \vec{r} \, \rho_b \, d^3r = -\int \vec{r} \left(\vec{\nabla} \cdot \vec{P}\right) d^3r =$$

$$= -\int r_i \frac{\partial P_j}{\partial r_j} d^3r = -\left\{ \int \frac{\partial}{\partial r_j}\left(r_i P_j\right) dV - \int \underbrace{\left(\frac{\partial r_i}{\partial r_j}\right)}_{\delta_{ij}} P_j \, dV \right\} = -\underbrace{\oint r_i P_j \, dS_j}_{\emptyset} + \int P_i \, dV$$

$$\Rightarrow \int \vec{P} \, dV = \int \vec{r} \, \rho_b \, dV$$

$\Rightarrow$ polarization $\vec{P}$ = dipole moment of unit volume

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho_b}{\varepsilon_0} = -\frac{1}{\varepsilon_0}\vec{\nabla}\cdot\vec{P} \Rightarrow \vec{\nabla}\cdot\left(\vec{E} + \frac{1}{\varepsilon_0}\vec{P}\right) = 0$$

$\Rightarrow$ Introduce <u>displacement vector</u> $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$$\vec{\nabla}\cdot\vec{D} = 0 \quad (\text{if } \rho_f = 0)$$

$$\vec{\nabla}\cdot\vec{D} = \rho_f \quad \text{in general.}$$

Continuity eqn: $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0$, $\rho_b = -\vec{\nabla}\cdot\vec{P}$

$$\Rightarrow \frac{\partial}{\partial t}\vec{\nabla}\cdot\vec{P} = \vec{\nabla}\cdot\vec{J}_b \Rightarrow \vec{\nabla}\cdot\left(\vec{J}_b - \frac{\partial \vec{P}}{\partial t}\right) = 0$$

Can write $\vec{J}_b = \frac{\partial \vec{P}}{\partial t}$? $\vec{J}_b$ is defined up to $\vec{\nabla}\times\vec{M}$!

Not quite: $\Rightarrow \vec{J}_b = \frac{\partial \vec{P}}{\partial t} + \vec{\nabla}\times\vec{M}$

$$\left[\begin{array}{l} \text{Since } \vec{\nabla}\cdot\left(\vec{\nabla}\times\vec{A}\right) = 0 \text{ for any } \vec{A} \\ \vec{\nabla}\times\left(\vec{\nabla} f\right) = 0 \text{ for any } f \end{array}\right]$$

Meaning of $\vec{M}$? <u>Magnetization</u>!

$$\begin{cases} \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{B} = \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J} \end{cases}$$

<u>Steady state</u> (time independent) $\frac{\partial \vec{E}}{\partial t} = 0$

Need to distinguish <u>free current</u> $\vec{J}_f$ from that of <u>bound charges</u> $\vec{J}_b$ (which is always present due to atomic scale motion)

For bound current $\oint \vec{J}_b \cdot d\vec{S} = 0$
over full cross-section of the body

(For conductors, $\oint \vec{J}_f \cdot d\vec{S} \neq 0$ in general! = net flow of charge)

can write $\Rightarrow$ $\vec{J}_b = \vec{\nabla} \times \vec{M}$ $\quad \left(J_{b\,i} = \varepsilon_{ijk} \frac{\partial}{\partial r_j} M_k \right)$

<u>Stokes' th:</u> $\int (\vec{\nabla} \times \vec{M}) \cdot d\vec{S} = \oint \vec{M} \cdot d\vec{l} = 0$

over cross-section — line integral along the boundary

choosing ~~bou~~ boundary outside the body where $\vec{M} = 0$!

i-th component of <u>magnetic dipole moment</u> of a medium

$$\vec{m} = \frac{1}{2} \vec{r} \times \vec{J} = \frac{1}{2} \vec{r} \times \rho \vec{v}$$

$$\left(\text{or, } \vec{m} = \frac{1}{2} \int \vec{r}\,' \times \vec{J}(\vec{r}\,') \, d^3 \vec{r}\,' \right)$$

$$\frac{\partial}{\partial r_\ell} (r_j M_m) - M_m \frac{\partial r_j}{\partial r_\ell}$$

$$\int m_i \, d^3\bar{r} = \frac{1}{2} \varepsilon_{ijk} \int r_j J_k \, d^3\bar{r} = \frac{1}{2} \underbrace{\varepsilon_{ijk} \varepsilon_{k\ell m}}_{\substack{\varepsilon_{ijk}\varepsilon_{\ell m k} \\ \parallel \\ \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}}} \int r_j \frac{\partial}{\partial r_\ell} M_m \, d^3\bar{r}$$

$$= \frac{1}{2} (\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}) \left\{ \underbrace{\int r_j M_m \, d^2 r_{n \neq \ell} \Big|^{r_\ell \in S}}_{\emptyset \text{ since } \vec{M} = 0 \text{ outside the medium}} - \int M_m \underbrace{\frac{\partial r_j}{\partial r_\ell}}_{\delta_{j\ell}} d^3\bar{r} \right\}$$

$$\int M_i \, d^3\bar{r} = \frac{1}{2} \sum_{jml=1}^{3} \left(\delta_{im}\delta_{jl} - \delta_{il}\delta_{jm}\right) \delta_{jl} \int M_m \, d^3\bar{r}$$

$$= \frac{1}{2} \sum_{l=1}^{3} \int M_i \, d^3\vec{r} - \frac{1}{2} \underbrace{\sum_{\substack{l=i \\ j=m \\ j=l}} \delta_{il}\delta_{jm}\delta_{jl}}_{} \int M_i \, d^3\bar{r} \rightarrow$$

$$\sum_{jm} \int \underset{\delta_{il}}{\underset{\downarrow}{M_l}} \, d^3r \; \underset{m=j}{\underset{\downarrow}{\delta_{jm}}} \; \underset{j=l}{\underset{\downarrow}{\delta_{jl}}} = \int M_i \, d^3\bar{r}$$

$\Rightarrow$ the sum $\sum_{jml=1}^{3}$ reduces to single term, $j=i$

$$\rightarrow \frac{1}{2}(3-1)\int M_i \, d^3\vec{r} = \int M_i \, d^3\vec{r}$$

$$\Rightarrow \int \vec{M} \, d^3\vec{r} = \frac{1}{2} \int \vec{r} \times \vec{J}_b(\bar{r}) \, d^3\vec{r} = \sum_V \vec{m}_n$$

Magnetization $\vec{M}$ = density of magnetic dipole moment $\vec{m}$.

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_b = \mu_0 \vec{\nabla} \times \vec{M}$$

$$\vec{\nabla} \times \left(\frac{1}{\mu_0}\vec{B} - \vec{M}\right) = 0 \Rightarrow \boxed{\text{magnetic field } \vec{H} = \frac{1}{\mu_0}\vec{B} - M}$$

$\boxed{\vec{B} = \text{magnetic induction}}$

$\vec{\nabla} \times \vec{H} = 0$ in a steady state.

Altogether: $\vec{J}_b = \frac{\partial \vec{P}}{\partial t} + \vec{\nabla} \times \vec{M}$

$$\rho_b = -\vec{\nabla} \cdot \vec{P}$$

(1) $\vec{\nabla} \cdot \left(\vec{E} + \frac{1}{\varepsilon_0} \vec{P} \right) = \frac{\rho_f}{\varepsilon_0} \Rightarrow \vec{\nabla} \cdot \left(\varepsilon_0 \vec{E} + \vec{P} \right) = \rho_f$

(1') $\vec{\nabla} \cdot \vec{D} = \rho_f$

(3) $\vec{\nabla} \cdot \vec{B} = 0$

(4) $\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$

} homogeneous Maxwell eqns retain their form

(2) $\vec{\nabla} \times \vec{B} - \varepsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t} = \mu_0 \left(\vec{J}_f + \frac{\partial \vec{P}}{\partial t} + \vec{\nabla} \times \vec{M} \right)$

$$\vec{\nabla} \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) - \varepsilon_0 \frac{\partial \vec{E}}{\partial t} - \frac{\partial \vec{P}}{\partial t} = \vec{J}_f$$

(2') $\vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{J}_f$

$$\begin{cases} \vec{\nabla} \cdot \vec{B} = 0 \\ \vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \\ \vec{\nabla} \cdot \vec{D} = \rho_f \\ \vec{\nabla} \times \vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{J}_f \end{cases}$$

$\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$

$\vec{H}$ = mag. field

$\vec{B}$ = mag. induction

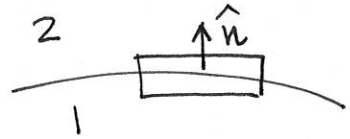
Constitutive relations: ~~need~~ to close Maxwell's eqns in the medium, we need to know

Linear medium

$$\left.\begin{array}{l} \vec{P} = \vec{P}(\vec{E}, \vec{H}) \\ \vec{M} = \vec{M}(\vec{E}, \vec{H}) \end{array}\right\} \xrightarrow{\text{linear medium!}} \left\{\begin{array}{l} \vec{P} = \varepsilon_0 \chi_e \vec{E} \\ \vec{M} = \chi_m \vec{H} \end{array}\right.$$

Note: $\vec{J}_f$ determines $\vec{H}$, while we usually do not control ρ_f rather $\rightarrow$ el. potential $\Phi \Rightarrow \vec{E} = -\vec{\nabla} \Phi$

Boundary conditions



(1) $$\int d^3r\,(\vec{\nabla}\cdot\vec{D}) = \int d\vec{S}\cdot\vec{D} = (\vec{D}_2-\vec{D}_1)\cdot\hat{n} = \int \rho_f\, d^3r$$

$$= \begin{cases} 0 & \text{for continuous charge dist.} \\ \sigma_f & \text{if there is a surface charge density} \end{cases}$$

i.e. $\rho_f = \rho_{f,\,\text{regular}} + \sigma_f\,\delta(z)$

$$\vec{D} = \varepsilon_0\vec{E} + \vec{P} \Rightarrow (\varepsilon_0\vec{E}_2 + \vec{P}_2 - \varepsilon_0\vec{E}_1 - \vec{P}_1)\cdot\hat{n} =$$

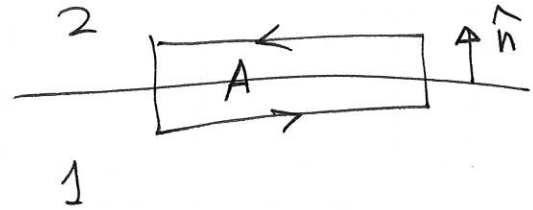
$$= \varepsilon_0(\vec{E}_2-\vec{E}_1)\cdot\hat{n} + (\vec{P}_2-\vec{P}_1)\cdot\hat{n}$$

$$(\vec{E}_2-\vec{E}_1)\cdot\hat{n} = \frac{\sigma_f + (\vec{P}_1-\vec{P}_2)\cdot\hat{n}}{\varepsilon_0} = \frac{\sigma_f+\sigma_b}{\varepsilon_0}$$

(If "2" is vacuum, then $\vec{P}_2 = 0$)

$$\sigma_b = \vec{P}_1\cdot\hat{n} - \vec{P}_2\cdot\hat{n}$$ bound surface-charge density

(2)



$$\int(\vec{\nabla}\times\vec{E})\cdot d\vec{A} = \oint \vec{E}\cdot d\vec{l} = -\int \frac{\partial\vec{B}}{\partial t}\cdot d\vec{A} \longrightarrow 0$$

when area $\vec{A}\to 0$.

$$E_{2\parallel} - E_{1\parallel} = 0$$ where "$\parallel$" is any direction in plane $\perp\hat{n}$

$$(\vec{E}_2-\vec{E}_1)\times\hat{n} = 0$$ ← hence

$$\Rightarrow \quad (\vec{D}_2 - \vec{D}_1)\times\hat{n} = \varepsilon_0 \underbrace{(\vec{E}_2 - \vec{E}_1)\times\hat{n}}_{\phi} + (\vec{P}_2 - \vec{P}_1)\times\hat{n}$$

or, $\quad D_{2,\parallel} - D_{1,\parallel} = P_{2,\parallel} - P_{1,\parallel}$.

Important:
$\vec{\nabla}\times\vec{D} = \varepsilon_0 \vec{\nabla}\times\vec{E} + \vec{\nabla}\times\vec{P} \neq 0$
generally

(3) $$\int (\vec{\nabla}\cdot\vec{B})\, dV = \int \vec{B}\cdot d\vec{S} = 0 \quad \text{always.}$$

$$(\vec{B}_2 - \vec{B}_1)\cdot\hat{n} = 0$$

$$\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M} \Rightarrow (\vec{H}_2 - \vec{H}_1)\cdot\hat{n} = \frac{1}{\mu_0}\underbrace{(\vec{B}_2 - \vec{B}_1)}_{\phi}\cdot\hat{n} - (\vec{M}_2 - \vec{M}_1)\cdot\hat{n}$$

$$(\vec{H}_2 - \vec{H}_1)\cdot\hat{n} = -(\vec{M}_2 - \vec{M}_1)\cdot\hat{n} \quad \left(\Rightarrow \vec{M}_1\cdot\hat{n} \text{ if } \vec{M}_2 = 0 \text{ outside}\right)$$

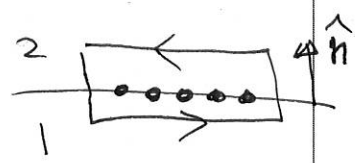
[Thus, while $\vec{\nabla}\cdot\vec{B} = 0 \rightarrow \vec{\nabla}\cdot(\vec{H} + \vec{M}) = 0$

$$\vec{\nabla}\cdot\vec{H} = -\vec{\nabla}\cdot\vec{M}$$]

(4) $$\int (\vec{\nabla}\times\vec{H})\cdot d\vec{A} = \underbrace{\int \frac{\partial\vec{D}}{\partial t}\cdot d\vec{A}}_{\substack{\phi \\ \text{when area}\to 0}} + \underbrace{\int \vec{J}_f\cdot d\vec{A}}_{\substack{\text{current through} \\ \text{the loop's area } A \\ \parallel \\ \vec{K}_f\times\hat{n}}}$$

$$\int (\vec{\nabla}\times\vec{H})\cdot d\vec{A} = \int \vec{H}\cdot d\vec{l}$$

$$H_{2,\parallel} - H_{1,\parallel} = \vec{K}_f \times \hat{n}$$



$$\cancel{\Rightarrow \frac{1}{\mu_0}(B_{2,\parallel} - B_{1,\parallel}) = \vec{M}_{2,\parallel} - \vec{M}_{1,\parallel}}$$

$$\cancel{(\vec{H}_2 - \vec{H}_1) \times \hat{n} = \vec{K}_f \times \hat{n} \times \hat{n} = \hat{n}(\vec{K}_f \cdot \hat{n})}$$

$$\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \hat{n} \times \hat{K}_f \times \hat{n} = \vec{K}_f(\hat{n}\cdot\hat{n}) - \hat{n}(\underbrace{\vec{K}_f \cdot \hat{n}}_{0})$$

$$\Rightarrow \quad (\vec{H}_2 - \vec{H}_1) \times \hat{n} = -\vec{K}_f$$

$$\Rightarrow \quad \frac{1}{\mu_0}(\vec{B}_2 - \vec{B}_1) \times \hat{n} - (\vec{M}_2 - \vec{M}_1) \times \hat{n} = -\vec{K}_f$$

$$(\vec{B}_2 - \vec{B}_1) \times \hat{n} = -\mu_0 \left[\underbrace{(\vec{M}_1 - \vec{M}_2) \times \hat{n}}_{} + \vec{K}_f \right]$$

$\vec{K}_b$ = effective surface current density of bound charges

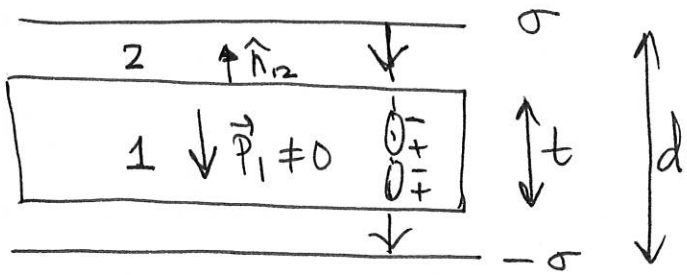
$$\vec{K}_b = \vec{M}_1 \times \hat{n} = \vec{M}_1 \times \hat{n}$$

if "2" = vacuum

"1" = media

Example: dielectric slab inside capacitor

$\vec{P}_2 = 0$



$\hat{n}_{12}$ = normal from "1" to "2".

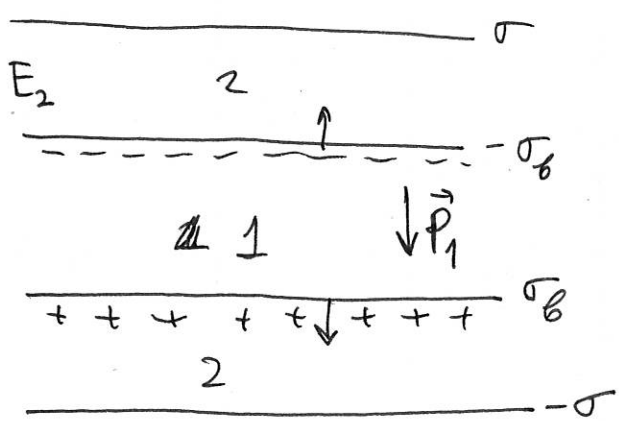
$$\left(\vec{E}_2 - \vec{E}_1\right)\cdot \hat{n}_{12} = \frac{\sigma_f + \vec{P}_1 \cdot \hat{n}}{\varepsilon_0}$$

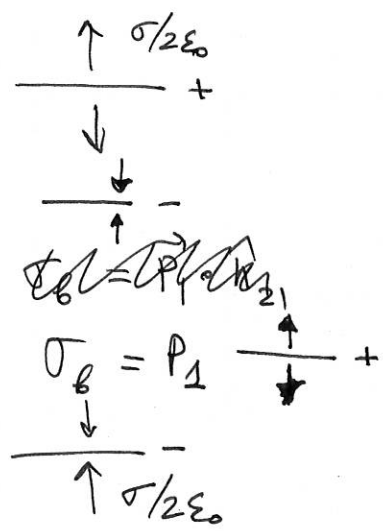
$$-\left(E_2 - E_1\right) = -\frac{P_1}{\varepsilon_0} \Rightarrow E_1 = \underset{\frac{\sigma}{\varepsilon_0}}{\underset{\parallel}{E_2}} - \frac{P_1}{\varepsilon_0} = \frac{\sigma - P_1}{\varepsilon_0}$$

$$\Rightarrow V = E_2(d-t) + E_1 t = \frac{\sigma(d-t) + (\sigma - P_1)t}{\varepsilon_0}$$

$$= \frac{\sigma d - P_1 t}{\varepsilon_0}.$$

Equivalently:





$$E_2 = \frac{\sigma}{2\varepsilon_0}\cdot 2$$

$$C = \frac{Q}{V}$$

$$E_1 = \frac{\sigma}{\varepsilon_0} - \frac{\sigma_b}{\varepsilon_0} = \frac{\sigma - P_1}{\varepsilon_0}$$

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$, $\vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \varepsilon_0 (1+\chi_e) \vec{E} = \varepsilon \vec{E}$

for linear dielectrics
(also, $\varepsilon = \varepsilon_0 \varepsilon_r$)

$$1+\chi_e = \varepsilon_r$$

$$\vec{P} = \varepsilon_0 (\varepsilon_r - 1) \vec{E}$$

$$E_1 = \frac{\sigma - P_1}{\varepsilon_0} = \frac{\sigma}{\varepsilon_0} - (\varepsilon_r - 1) E$$